# Why MNA

Modified nodal analysis builds G and b by stamping each element into a shared matrix

# The idea

- GOLDEN: mna size 3 , unknowns v1, v2, branch current i_Vin

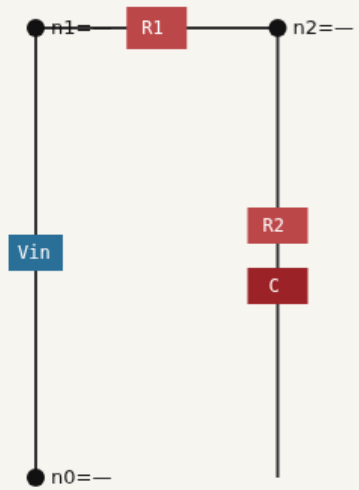
# Stamp R

MNA STAMPS

**Stamp R**

Step 1 / 5 · r · module01-03-mna-stamps

elements=5




**Explanation**

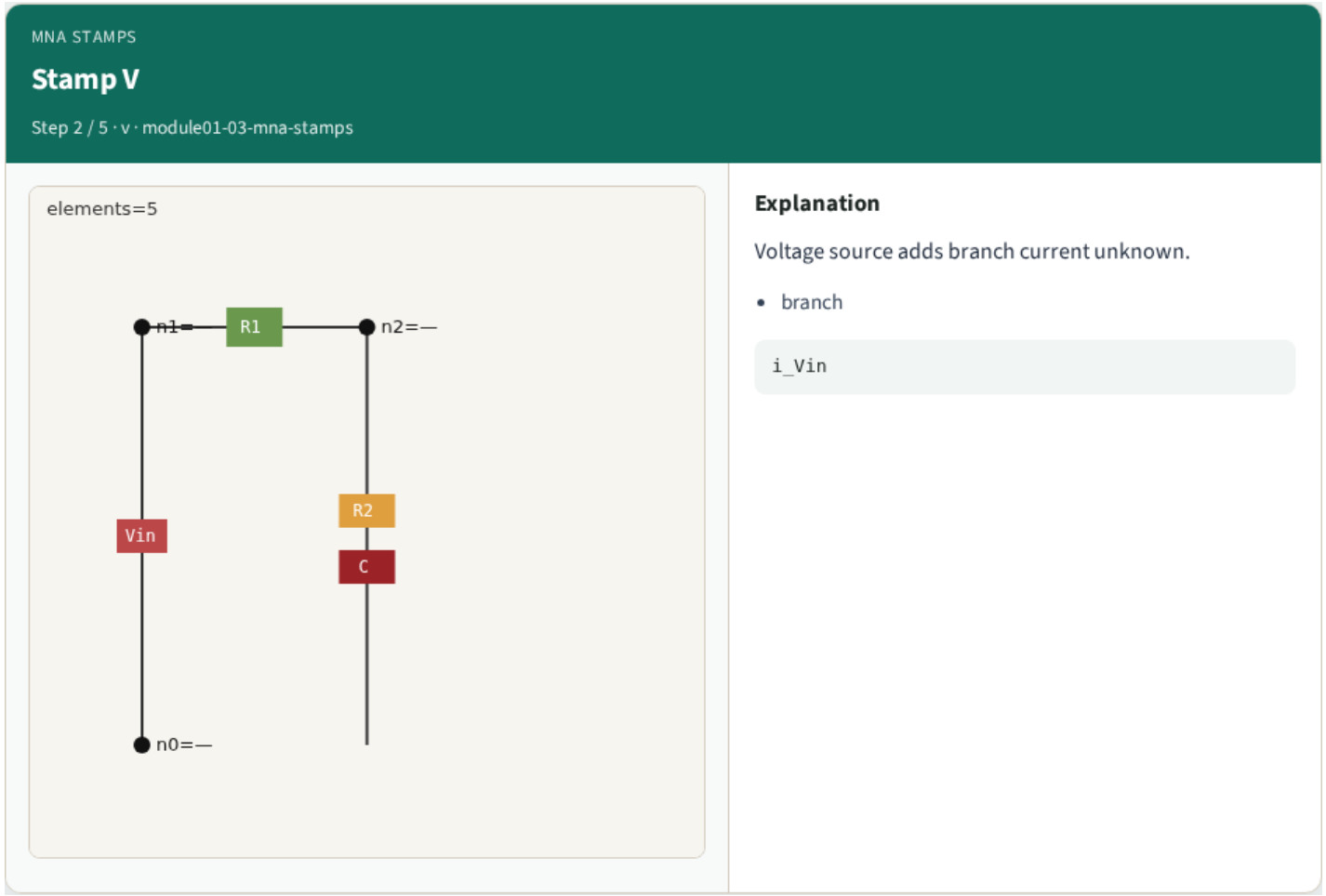
Conductance 1/R into G.

- stamp

```
1/R
```

Stamp R

# Stamp V




Stamp V

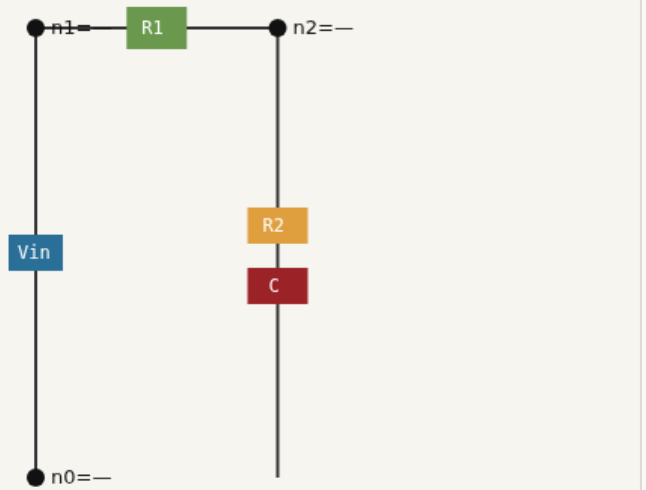
# Matrix G

MNA STAMPS

## Matrix G

Step 3 / 5 · g · module01-03-mna-stamps

size=3




**Explanation**

Build dense G for the toy.

- G

3×3

Matrix G

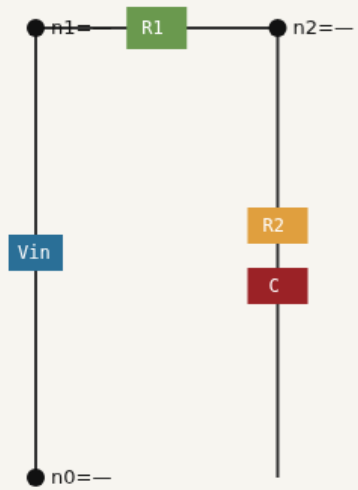
# RHS b

MNA STAMPS

## RHS b

Step 4 / 5 · b · module01-03-mna-stamps

b[Vin]=5




**Explanation**

RHS holds source values.

- b

```
Vin.dc
```

RHS b

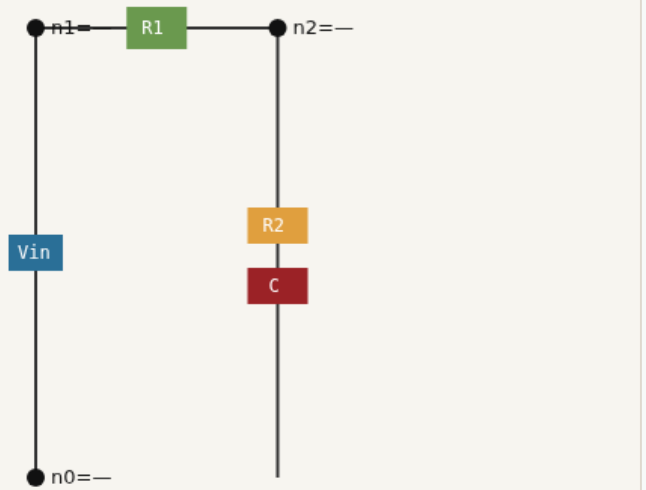
# Size 3

MNA STAMPS

**Size 3**

Step 5 / 5 · sz · module01-03-mna-stamps

size=3




**Explanation**

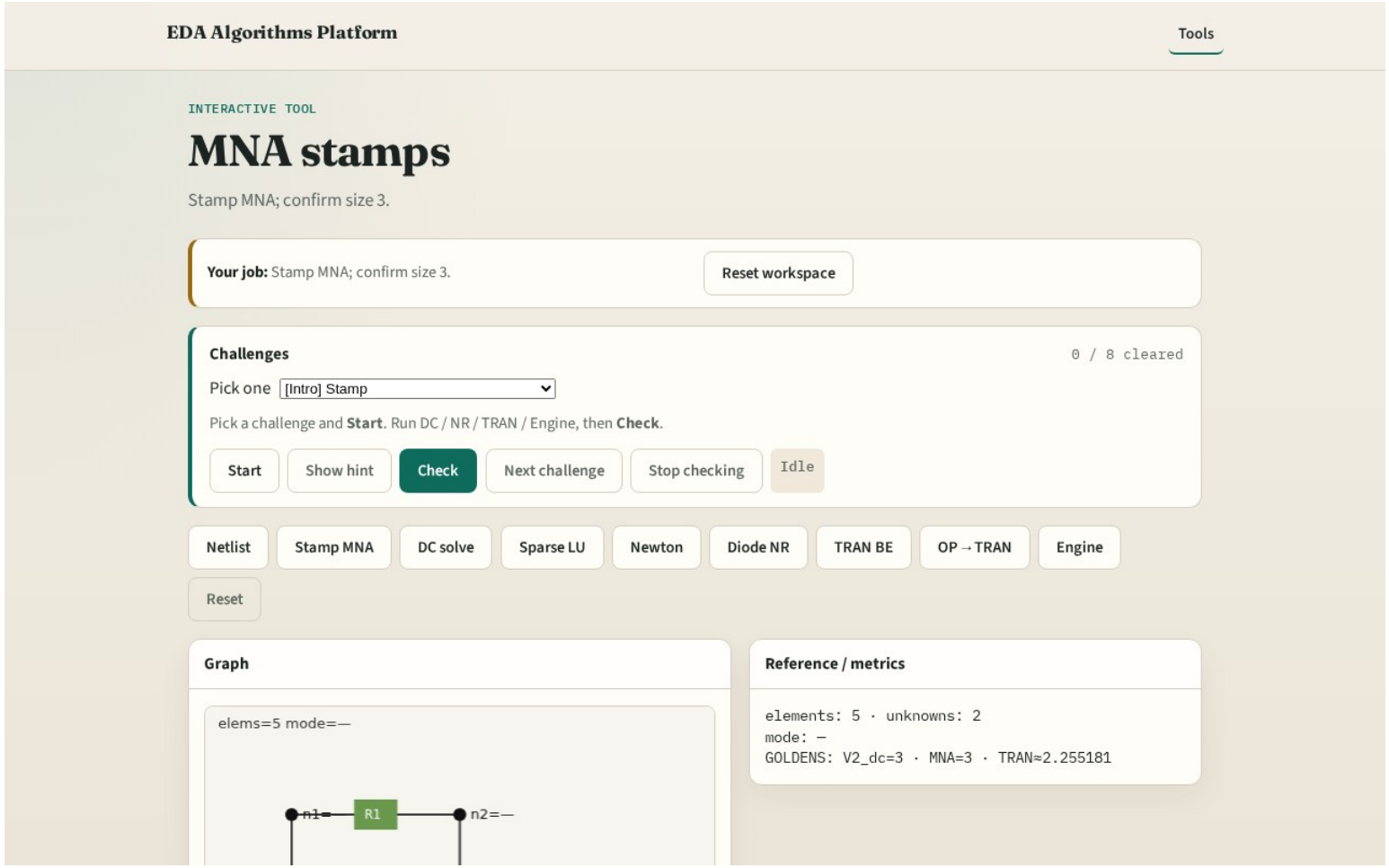
GOLDEN MNA size=3.

- GOLDEN

```
3
```

Size 3

# Browser lab track




Browser lab starter

# Implement track

- Call build_mna_linear_dc(); assert size==3

# Pitfalls

- Forgetting voltage-source branch currents
- Stamping C or D in linear DC MNA

# Your turn

- Next: solve linear DC for V2=3